# NEET-UG 2026 — Physics Categorization

Each of the 45 Physics questions in the NEET-UG 2026 paper, mapped to its corresponding Unit / Subtopic in the official NEET-UG 2026 syllabus.

***Sources:*** *NEET-UG 2026 syllabus (NMC notification, Notice_20260108180635.pdf, Physics — 20 units) · Question paper + answer key + line-by-line explanations (questions on pp. 2–86, explanations on pp. 87–277).*

## Distribution across the 20 syllabus units

| Unit | Syllabus Unit | Q count | Question Numbers |
|---|---|---|---|
| 1 | Physics and Measurement | 2 | Q8, Q35 |
| 2 | Kinematics | 2 | Q1, Q29 |
| 3 | Laws of Motion | 2 | Q5, Q7 |
| 4 | Work, Energy and Power | 1 | Q3 |
| 5 | Rotational Motion | 2 | Q27, Q30 |
| 6 | Gravitation | 1 | Q18 |
| 7 | Properties of Solids and Liquids | 2 | Q10, Q45 |
| 8 | Thermodynamics | 1 | Q11 |
| 9 | Kinetic Theory of Gases | 1 | Q23 |
| 10 | Oscillations and Waves | 4 | Q6, Q25, Q33, Q38 |
| 11 | Electrostatics | 3 | Q20, Q37, Q41 |
| 12 | Current Electricity | 4 | Q22, Q28, Q39, Q43 |
| 13 | Magnetic Effects of Current and Magnetism | 3 | Q12, Q16, Q31 |
| 14 | Electromagnetic Induction and Alternating Currents | 3 | Q19, Q26, Q40 |
| 15 | Electromagnetic Waves | 1 | Q42 |
| 16 | Optics | 4 | Q13, Q15, Q21, Q36 |
| 17 | Dual Nature of Matter and Radiation | 2 | Q2, Q4 |
| 18 | Atoms and Nuclei | 3 | Q17, Q32, Q44 |
| 19 | Electronic Devices | 3 | Q9, Q14, Q24 |
| 20 | Experimental Skills | 1 | Q34 |
| **Total** | | **45** | |

### Class-level split (NCERT)

Approximate, based on the dominant NCERT chapter:

- **Class XI** topics (Units 1–10): ~18 questions
- **Class XII** topics (Units 11–19): ~25 questions

## Per-question categorization

| Q# | Ans | Syllabus Unit | Subtopic | What the question tests (from the PDF explanation) |
|---|---|---|---|---|
| 1 | D | 2 — Kinematics | Position/velocity-time graphs; uniformly accelerated motion under gravity | Identify correct v-t plot for a ball thrown vertically up and returning — velocity is +ve, drops to 0 at apex, then becomes –ve. |
| 2 | D | 17 — Dual Nature of Matter & Radiation | Photoelectric effect; threshold wavelength | Condition for photoelectric emission: photon energy $hc/\lambda \geq \phi$, i.e. $\lambda \leq hc/\phi$. |
| 3 | C | 4 — Work, Energy and Power | Power; work against gravity | Crane lifts 1000 kg by 20 m in 10 s; P = mgh / t. |
| 4 | D | 17 — Dual Nature of Matter & Radiation | Energy of photon ($E=h\nu$), de Broglie wavelength ($\lambda=h/p$), wave/particle nature | List I ($E=h\nu$, Interference, $\lambda=h/p$, Compton) matched to List II (energy of photon, particle nature, de Broglie wavelength, wave nature). |
| 5 | B | 3 — Laws of Motion | Newton's 2nd law; resultant of perpendicular forces | Two perpendicular forces (8 N, 6 N) on 5 kg body → resultant 10 N → a = 2 $m/s^2$. |
| 6 | B | 10 — Oscillations and Waves | Energy in SHM; simple pendulum | Pendulum total energy 0.02 J → speed at mean position from KE = $\frac{1}{2}mv^2$. |
| 7 | A | 3 — Laws of Motion | Static friction; equilibrium on an accelerating surface | Max trolley acceleration so box remains stationary → $a_{max} = \mu_s \cdot g$. |
| 8 | C | 1 — Physics and Measurement | System of units; unit conversion | Sun-Earth distance with c = 1 (new unit system); distance = c × t = 400 light-seconds = 400 new-units. |
| 9 | B | 19 — Electronic Devices | Semiconductor diode — half-wave rectification | Voltage across a single diode in series with R fed by AC source — clipped sine (negative half blocked). |
| 10 | A | 7 — Properties of Solids and Liquids | Pressure due to a fluid column; Pascal's law | Max depth before submarine hits 100 atm absolute: $P_{abs} = P_{atm} + \rho gh$. |
| 11 | D | 8 — Thermodynamics | First law of thermodynamics | dU/dt = dQ/dt – dW/dt — rate of change of internal energy from heat in and work done. |
| 12 | C | 13 — Magnetic Effects of Current and Magnetism | Biot-Savart law — circular coil | B at centre of circular coil with N turns: B = μ■NI / (2r) → solve for I. |
| 13 | C | 16 — Optics | Wave optics — Young's double-slit (intensity, phase) | $I = I_{max} \cos^2(\phi/2)$; phase difference from given path difference. |
| 14 | C | 19 — Electronic Devices | Ideal diode circuit analysis | Multi-branch circuit with ideal diodes; identify forward-biased branches and apply KVL/KCL for net current I. |
| 15 | B | 16 — Optics | Refraction through thin lenses — concave (diverging) lens ray rules | Ray-tracing rules for a concave lens (parallel ray diverges; ray through optical centre is undeviated). |
| 16 | D | 13 — Magnetic Effects of Current and Magnetism | Moving coil galvanometer → ammeter (shunt) | Shunt $S = I_g G / (I - I_g)$ for G=100 Ω, $I_g$=1 mA, I=10 A. |
| 17 | B | 18 — Atoms and Nuclei | Bohr model — orbit radius, energy levels | First excited state (n=2) of H; r_n = n²a■ → r_2 = 4a■. |
| 18 | D | 6 — Gravitation | Gravitational PE; work done against gravity | Work to move mass between two radial distances from Earth's centre: W = ΔU = –GMm(1/r■ – 1/r■). |
| 19 | B | 14 — Electromagnetic Induction and Alternating Currents | LCR series circuit — resonance | f■ = 1 / (2π√(LC)) (independent of R). |
| 20 | D | 11 — Electrostatics | Capacitors — charge sharing & energy loss | Charged capacitor (200 pF at 100 V) connected to identical uncharged one → energy lost = ½ × initial energy. |
| 21 | B | 16 — Optics | Refraction through a prism — Snell's law | Equilateral prism, ray QR parallel to base → r = 30°; angle of deviation from i = 50°. |

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|---|---|---|---|---|
| **22** | **B** | 12 — Current Electricity | Metre/Wheatstone bridge — balance condition | Effect of interchanging galvanometer and cell positions on detected balance point. |
| **23** | **B** | 9 — Kinetic Theory of Gases | RMS speed; degrees of freedom / molar mass | $v_{rms} \propto 1/\sqrt{M}$; ratio for Argon (40) vs Chlorine (71). |
| **24** | **A** | 19 — Electronic Devices | p-n junction diode — forward bias / I-V characteristics | Statements on threshold/knee voltage and exponential current rise in forward bias. |
| **25** | **A** | 10 — Oscillations and Waves | Travelling wave — phase difference vs path difference | $y = 2.0 \cos 2\pi(10t – 0.0080x + 0.35)$; $\Delta\Phi = 2\pi \times 0.0080 \times \Delta x$. |
| **26** | **A** | 14 — Electromagnetic Induction and Alternating Currents | Faraday's / motional emf | emf = Blv with B=0.3 T, l=3 cm, v=2 cm/s → 1.8 × 10■■ V. |
| **27** | **D** | 5 — Rotational Motion | Moment of inertia; parallel axis theorem | Ring about a tangent in its plane: $I = \frac{1}{2}mR^2 + mR^2 = (3/2)mR^2$ with R from wire length $L = 2\pi R$. |
| **28** | **A** | 12 — Current Electricity | EMF, internal resistance, terminal voltage | $V_T = E – Ir$ for E=12 V, r=2 Ω, I=0.6 A. |
| **29** | **C** | 2 — Kinematics | Uniformly accelerated motion under gravity (reaction-time experiment) | Ruler falls from rest: $d = \frac{1}{2}gt^2$ — distance proportional to $t^2$. |
| **30** | **B** | 5 — Rotational Motion | Equations of rotational motion (angular kinematics) | Flywheel angular speed 600→1200 rpm; revolutions from $\theta = \omega_{avg} \times t$ (or ω² = ω■² + 2αθ). |
| **31** | **A** | 13 — Magnetic Effects of Current and Magnetism | Ampère's law — long straight current-carrying wire | B(r) for solid wire of radius a: $B \propto r$ for $r \le a$, $B \propto 1/r$ for $r \ge a$. |
| **32** | **B** | 18 — Atoms and Nuclei | Size of nucleus; mass number relations | $R \propto A^{(1/3)} \Rightarrow V \propto A$; identify true statements about nuclear volume and density. |
| **33** | **B** | 10 — Oscillations and Waves | Simple pendulum — time period | T = 2 s from 30 oscillations in 60 s; $L = gT^2/(4\pi^2)$. |
| **34** | **B** | 20 — Experimental Skills | Vernier calipers — least count | 20 VSD = 16 MSD, 1 MSD = 1 mm → LC = 1 MSD – 1 VSD = 0.2 mm. |
| **35** | **B** | 1 — Physics and Measurement | Significant figures; error propagation | Density of cube from mass (4 sig fig) and side (2 sig fig); answer limited to 2 sig fig. |
| **36** | **C** | 16 — Optics | Wave optics — interference & diffraction (conceptual) | Evaluate statements: energy redistribution (true), property of all waves (true/false). |
| **37** | **D** | 11 — Electrostatics | Capacitors in series/parallel; charge on each capacitor | Four 10 μF in series, parallel with 2.5 μF → equivalent C and individual charges. |
| **38** | **B** | 10 — Oscillations and Waves | Energy in SHM — KE vs time graph | KE non-negative, oscillates at 2× pendulum frequency, max at mean position, zero at extremes. |
| **39** | **D** | 12 — Current Electricity | Electric power (heating effect) | Heater rated 400 W at 220 V; new P at reduced V keeping R constant. |
| **40** | **B** | 14 — Electromagnetic Induction and Alternating Currents | Alternating current — instantaneous value, peak time | $I(t) = I_p \sin(\omega t)$; reach peak when $\omega t = \pi/2 \rightarrow t = T/4 = 1/(4f)$. |
| **41** | **A** | 11 — Electrostatics | Conductors in electrostatic equilibrium; surface field | Truth of statements: field inside conductor is 0 (true); surface E depends on σ (true). |
| **42** | **C** | 15 — Electromagnetic Waves | EM spectrum — production methods | Match microwave→klystron/magnetron, visible→atomic transitions, gamma→nuclear decay, IR→thermal radiation. |
| **43** | **D** | 12 — Current Electricity | Kirchhoff's laws; balanced bridge | Square loop made from 4 Ω wire (each side 1 Ω) with 2 Ω across diagonal, 2 V battery — solve for I. |

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| --- | --- | --- | --- | --- |
| **44** | **A** | 18 — Atoms and Nuclei | Nuclear density; R = R■A^(1/3) | Find mass number A from given nuclear density and R■ = 1.2 × 10■$^{1}$■ m. |
| **45** | **A** | 7 — Properties of Solids and Liquids | Elastic moduli — Young's modulus, compressibility, Poisson's ratio | Match physical property to defining formula ($Y = FL/A\Delta L$, $\beta = -(1/V)(\Delta V/\Delta P)$, etc.). |

## Notes on borderline / cross-listed items

A few questions sit at the boundary of two syllabus units. These are the choices made and why:

### Q14 (diode circuit)

Placed under Unit 19 (Electronic Devices) because the explanation hinges on identifying which diodes are forward-biased; KVL/KCL is mechanical and secondary. Cross-list: Unit 12.

### Q22 (metre bridge)

Kept under Unit 12 (Current Electricity) since the syllabus places the metre bridge there explicitly. Q43 also fits here (balanced bridge with Kirchhoff).

### Q29 (reaction-time / ruler fall)

Categorized as Unit 2 (Kinematics) — the experimental dressing is incidental; underlying physics is $d = \frac{1}{2}gt^2$. Could also be filed under Unit 20.

### Q33 (simple pendulum T)

Primary unit is Unit 10 — Oscillations & Waves ($T = 2\pi\sqrt{(L/g)}$). The framing is experimental (30 oscillations in 60 s), so it also touches Unit 20.

### Q34 (vernier least count)

Pure experimental skill → Unit 20; also implicitly Unit 1 (measurement).

### Q35 (density with sig figs)

Primarily about significant figures and error propagation → Unit 1.

### Q4 ($E=h\nu$, $\lambda=h/p$, Compton, interference)

Bridges Units 16 and 17. Placed under Unit 17 because three of four sub-items are about photons / matter waves; interference appears only as a misdirection in this matching question.

### Q45 (elastic moduli matching)

Placed under Unit 7 (Properties of Solids and Liquids) since Young's modulus, bulk modulus and compressibility live there.